**Design and Simulation of a Two-Stage BJT Amplifier**

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**Software Used:** LTspice

**1. Project Objective**

To design, calculate, and simulate a two-stage cascaded NPN Bipolar Junction Transistor (BJT) amplifier that meets specific voltage gain and biasing constraints, while achieving a stable and physically realistic mid-band frequency response.

**2. Design Specifications**

- **Supply Voltage :**12V
- **Target Voltage Gain - Stage 1:** 30 V/V
- **Target Voltage Gain - Stage 2 :** 20 V/V
- **Overall Target Voltage Gain :**600 V/V (approx. 55.5 dB)
- **Load Resistance :** 1.5kΩ
- **Biasing Constraints (for maximum unclipped swing):**
    - Collector Voltage = 0.5 Vdd(6V)
    - Emitter Voltage = 0.1 Vdd (1.2V)

**3. Circuit Topology**

The circuit utilizes two Common-Emitter (CE) amplifier stages cascaded in series.

- **Active Components:** The design uses the standard 2N2222 BJT model for both stages to ensure realistic simulation of parasitic capacitances.
- **Stage 1 & 2:** Provide voltage amplification with gains of 30 and 20, respectively, utilizing voltage divider biasing schemes.
- **Coupling & Bypassing:** AC coupling capacitors isolate the DC operating points of each stage. Emitter bypass capacitors short the AC signal to ground, preventing negative feedback and maximizing mid-band voltage gain.

**4. Component Values**

**4.1 Resistors (DC Biasing and Gain Setting)**

- **Stage 1:** R1 = 7.2kΩ, R2 = 1.44kΩ, Rc = 300Ω, Re1 = 3Ω, Re2 = 57Ω
- **Stage 2:** R1 = 3.6kΩ, R2 = 720Ω, Rc = 150Ω, Re1 = 3Ω, Re2 = 27Ω
- **Load:** Rl = 1.5kΩ

**4.2 Capacitors (Frequency Response Tuning)**

Capacitor values were selected to set the dominant low-frequency cutoff ($f_L$) to approximately 50 Hz:

- **Input Coupling (C7):** 47 µF
- **Interstage Coupling (C5):** 33 µF
- **Output Coupling (C1):** 10 µF
- **Stage 1 Emitter Bypass (C4):** 1000 µF
- **Stage 2 Emitter Bypass (C3):** 1000 µF

**5. Simulation Results (AC Analysis)**

An AC sweep from 1 Hz to 1 GHz was performed to evaluate the amplifier's frequency response. The resulting Bode plot demonstrates a textbook bandpass characteristic, validating the physical design of the circuit.

- **Mid-Band Gain:** The amplifier achieves a stable, flat mid-band voltage gain of approximately 60 dB. This amplification region is highly linear, spanning consistently from roughly 100 Hz up to 1 MHz.
- **Low-Frequency Response:** The lower -3 dB cutoff frequency is precisely positioned in the sub-100 Hz range. The curve rolls off smoothly below this point, confirming that the coupling and bypass capacitors are correctly sized to pass the desired signal frequencies without premature attenuation.
- **High-Frequency Response:** The upper -3 dB cutoff frequency occurs in the megahertz range (approximately 1 MHz to 5 MHz). The gain correctly and naturally decays at higher frequencies, accurately reflecting the bandwidth limitations imposed by the internal base-emitter and base-collector junction capacitances of the 2N2222 transistor model.
- **Effective Bandwidth:** The resulting bandwidth successfully covers a wide frequency range, demonstrating a stable design suitable for applications extending from the audio spectrum into the low-megahertz range.